Maneuver Prediction from Ego-Vehicle Camera Images

DATASCI 281 · Group 2 | Dawson · Matt · Nithya · Victor

Motivation & Task

THE TASK

Predict the ego-vehicle's upcoming maneuver from camera imagery alone (WOD-E2E panoramas).

straight | left-turn | right-turn

WHY IT MATTERS

Anticipating maneuvers from pixels is a core layer of the driving stack, and for **other road users** it is the only option. Ego sequences give us labeled ground truth to build and measure exactly that capability.

WHY IT'S HARD

- One camera stream — no lidar, radar, or map fusion.
- Classical features see a single frame of a motion event.
- Temporal features see only the short window before the trajectory changes.

ONE LAYER OF THE DRIVING STACK

Sensing
cameras · lidar · radar

Perception
detection · segmentation · tracking

Maneuver anticipation
THIS PROJECT — camera-only, 3-class

Prediction & planning
trajectories · behavior · route

Control
steering · acceleration

We isolate the highlighted layer — how far can a single camera view take it?

Dataset & the Pivot

Waymo Open Dataset End-to-End (WOD-E2E) · StandardE2E preprocessing

THE DATA

1,966

3-class sequences (train 2,037 + val 479 pooled, filtered)

142×384

8-camera panorama per frame (RGB)

5 s

future ego-trajectory → label (Waymo ClassifyTrack port)

70/15/15

frozen stratified split — 1,376 / 295 / 295, joined by seq id

CLASS BALANCE (pooled)

THE PIVOT — decisions along the way

1 5 classes → 3

Lane-changes have no present-frame signature (five independent null tests) and their labels conflate in-lane curve-following. Dropped — not folded into straight.

2 Waymo test unlabeled → pooled re-split

Future trajectories are withheld on the official test split, so train + validation were pooled and re-split 70/15/15, stratified, fixed seed.

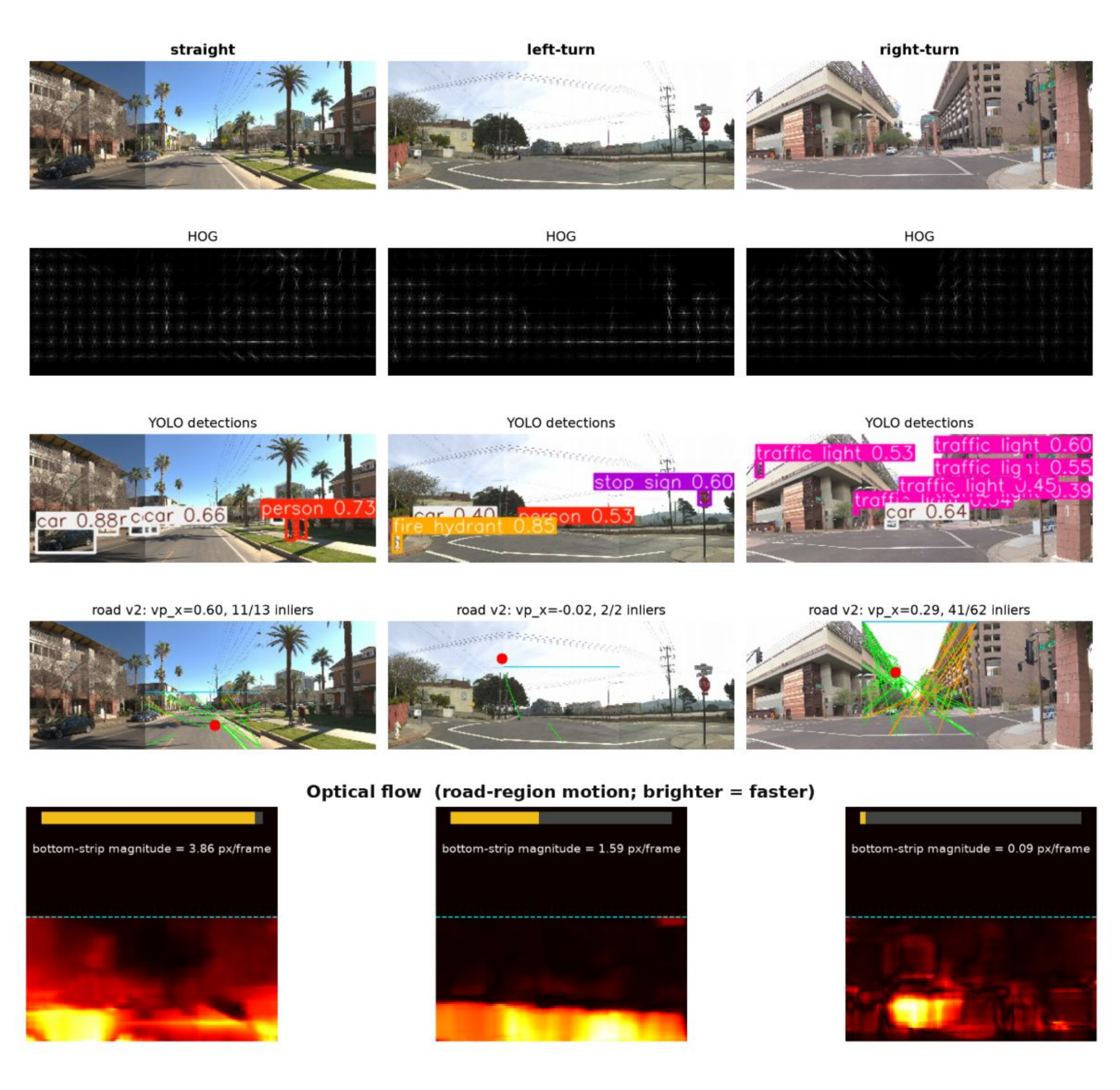
3 Keep night driving (30% of data)

Day-only training did not help on matched daytime test folds — night is harder, not broken. Reported as a split, not filtered out.

Feature Zoo

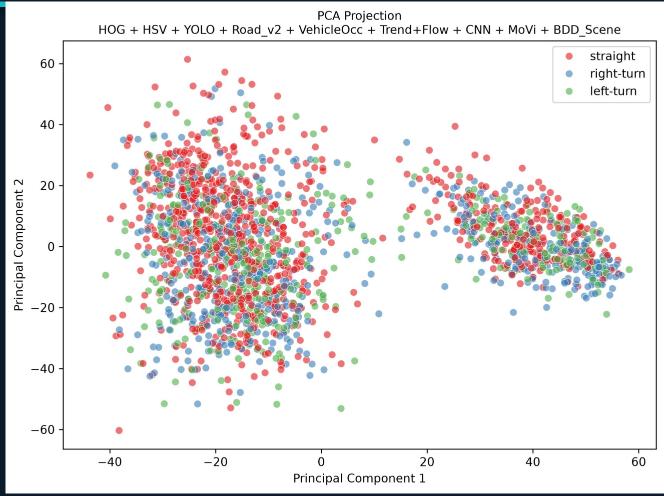
The features we engineered and benchmarked

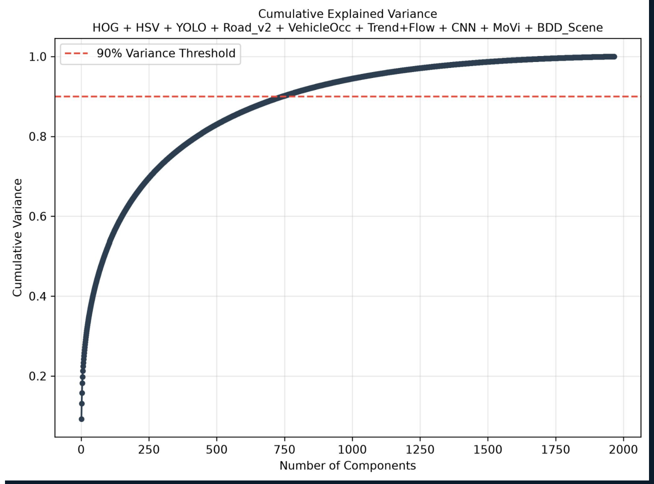
- Simple: HOG (edges, lane lines); HSV histogram (color, lighting)
- Geometry & objects: YOLO detections + position; vehicle occupancy (counts + spatial layout); Road_v2 (Hough lines + RANSAC vanishing point)
- Temporal: Trend + Optical Flow, how the scene moves in the frame before the maneuver
- Learned: 2D CNN penultimate, frozen MoViNet video embedding



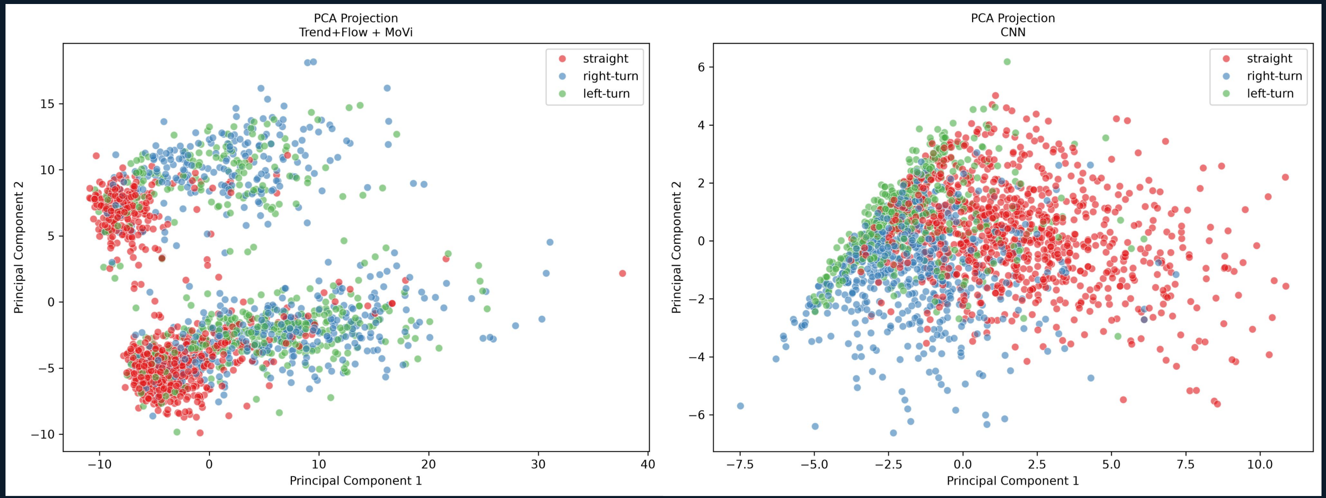
Feature Separability

- Over the entire set, we do not see clear separability
- Large subset of features are needed to explain variance





- Certain subsets show strong evidence of separability
- Limitation of classical features vs complex features from pre-trained models



Models & Protocol

Stage	Implementation	Guardrail
Split	70/15/15 stratified, fixed seed	Same sequence order for every feature set
Training	Fit only on the training fold	Weighting and augmentation from train only
Selection	Tune on validation macro-F1	Test never used for hyperparameters
CNN early stopping	Internal train-only validation fold	Shared validation remains comparable
Final reporting	Accuracy, macro-F1, per-class recall	Held-out test evaluated once

HYPERPARAMETER SEARCH

grid-searched · selected on validation macro-F1

SVM (rbf)	C	**1**	10
	γ	**scale**	
RF (300)	depth	**None**	20

LogReg baseline: C ∈ {0.1, 1, 10} → 0.1–1.0 by feature set

SELECTION

Best val macro-F1, hand-crafted combos only

→ **Trend+Flow / RF (val 0.611)**

Frozen embeddings (MoViNet, BDD): ablations

CNN features: ablation-only (val leakage)

FINAL MODELS

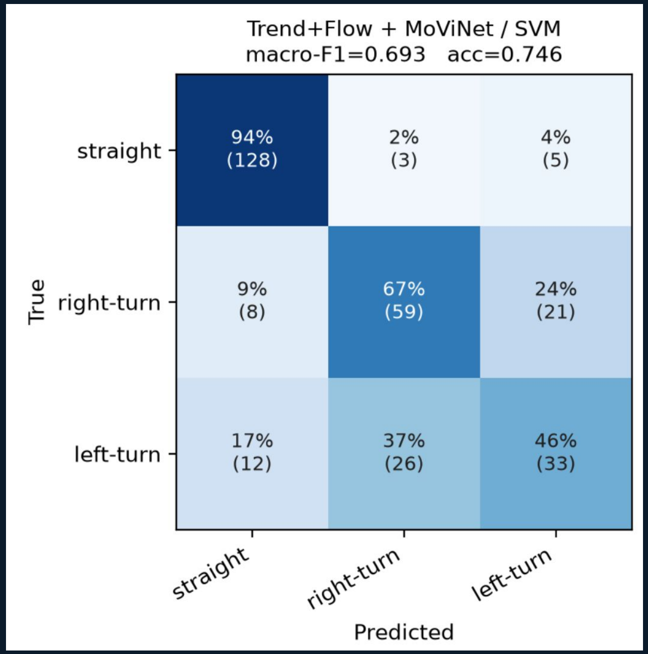
Classical: Trend+Flow / RF (300, depth None)

Stack: Trend+Flow + MoViNet / SVC (rbf, C=1)

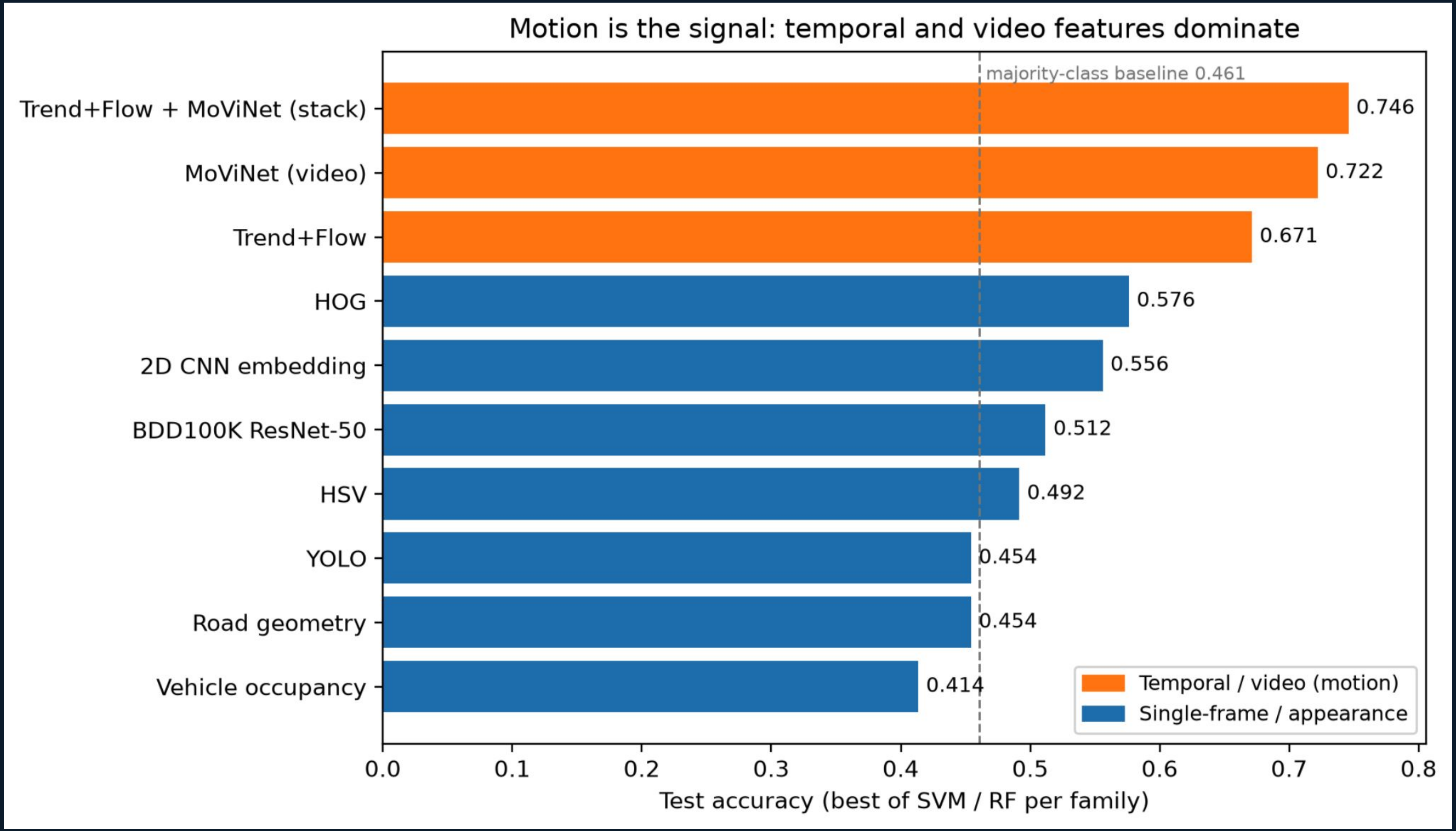
Results

Best: Trend+Flow + frozen MoViNet / SVM — 0.746 acc / 0.693 macro-F1

Pipeline	Accuracy	Macro-F1	Main signal
Majority baseline	0.461	0.210	None
BDD100K ResNet-50	0.512	0.441	Single frame, driving-pretrained
2D CNN	0.563	0.516	Single frame
Prior-only 3D CNN	0.624	0.563	Learned motion
Frozen MoViNet head	0.675	0.599	Video transfer
Trend+Flow / RF	0.671	0.607	Explicit motion
Trend+Flow + MoViNet / SVM	**0.746**	**0.693**	**Fused motion**



Motion is the Signal



Efficiency vs. Accuracy

Pipeline	Feature extraction	Model training	Accuracy	Macro-F1	Measured time
2D CNN	Medium	High	0.563	0.516	321 seconds
3D CNN from scratch	High	High	0.624	0.563	1606 seconds
Frozen MoViNet head	High 1x; cached	Low	0.675	0.599	hours+
Cached Trend+Flow + MoViNet / SVM	**High 1x; cached**	**Lowest**	**0.746**	**0.693**	0.20 seconds

Best downstream model was cheapest to train after caching—not the cheapest end-to-end feature pipeline.

What We Learned

Motion is the signal

22 hand-crafted temporal dimensions (Trend+Flow) beat every single-frame family — classical or learned. 0.671 acc / 0.607 F1 vs ≤ 0.576 for any appearance feature.

Video transfer complements explicit motion

Frozen MoViNet embeddings + Trend+Flow under an SVM: 0.746 acc / 0.693 F1 — +0.086 F1 over the classical best. Learned and hand-crafted motion are complementary.

Domain pretraining ≠ motion

A driving-pretrained single-frame embedding (BDD100K ResNet-50) lands in the appearance cluster and costs −0.14 F1 when stacked. The temporal dimension is what matters.

Left-turn is the hard class — everywhere

Recall 0.25–0.46 across every model. It is the rarest class (24%) with the least prior context in the data — a class asymmetry, not a modeling choice.

Bonus lesson — validation selection is noisy at n=295: unrestricted val selection picked a feature stack that collapsed on test (0.623 val ⟶ 0.510). Guardrails: select among compact hand-crafted combos; treat frozen embeddings as ablations.

Limitations & Future Work

Limitation	Why it matters	Next step
Single 70/15/15 split	Metric variance is unknown	Repeat seeds or grouped cross-validation
1,966 labeled sequences	Deep video models can overfit	Add data + class-balanced augmentation; no horizontal flips
Future-trajectory labels	Observed maneuver is not identical to driver intent	Test label horizon and intent-aligned labels
Official test labels unavailable	Custom split is not an official benchmark	Publish split seed and sequence IDs
CPU-only video extraction	Slow iteration on learned video features	GPU fine-tuning or a larger video-pretrained backbone
Ego vehicle only	Reading manuevers from pixels matters most for *other* road users (no sensor, lidar data for other vehicles)	Same pipeline on observed vehicles: track + crop each agent, ego-motion-compensated flow, ClassifyTrack labels from WOMD agent tracks

EXTRA WORK

Completed prior-only 3D CNN + Kinetics-pretrained MoViNet; add remaining team commitments here.

Q&A

- Questions?